Northeastern University

We built a reconciliation data platform — then used it to test how AI should review code

A real client needed reconciliation data made legible. We shipped a production platform for the **Indigenomics Institute** — and its own **30 merged pull requests** became the industrial dataset for a pre-registered study of multi-agent AI code review.

Research companion: "Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study"

En-Ping Su · Tong Wu · Shiting Huang · Mengshan Li
Khoury College of Computer Sciences, Northeastern University · Vancouver · Advisor: Yvonne Coady · Industry Partner: Indigenomics Institute (Carol Anne Hilton, Founder)

RESEARCH CONTRIBUTION

More AI reviewers don't help — decorrelation does. Agreement predicts truth only when the reviewers are independent.

89% 3 independent families agree → matches ground truth

54% one model, three runs → mostly noise

01 THE PROJECT — RAP DATA PORTAL

PROBLEM

Companies publish reconciliation commitments. Legal precedents sit scattered across court records. There is no searchable, verifiable, unified view — so nobody can track progress.

↓

OUR SOLUTION

- Searchable RAP commitments
- Legal precedent database
- AI extraction from PDFs
- Human review workflow
- AWS serverless hosting
- Authentication & roles
- Progress tracking
- Coverage analytics

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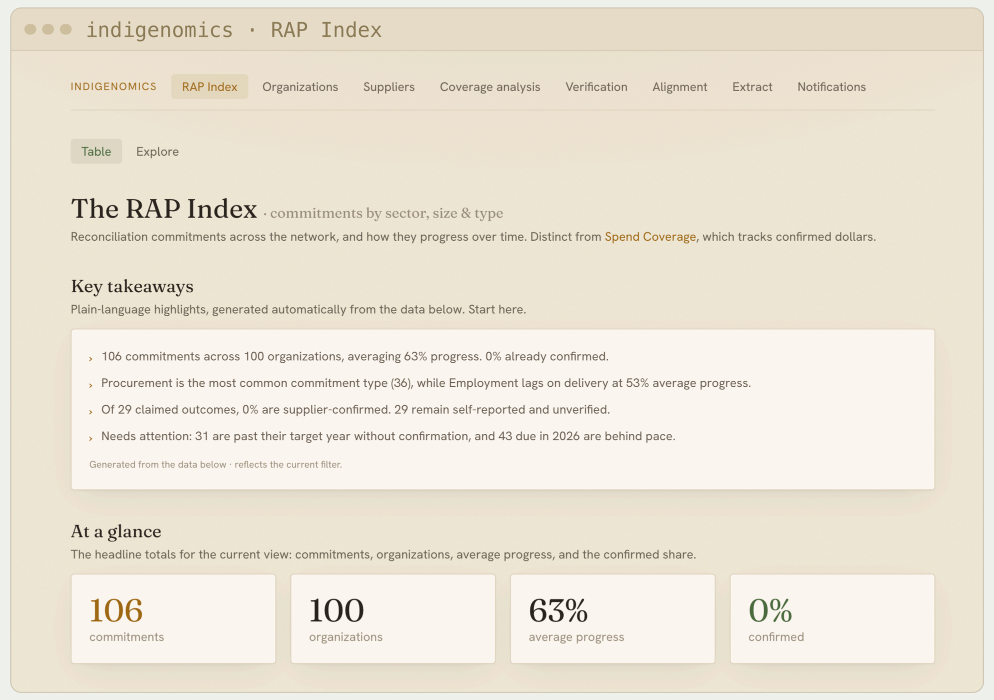
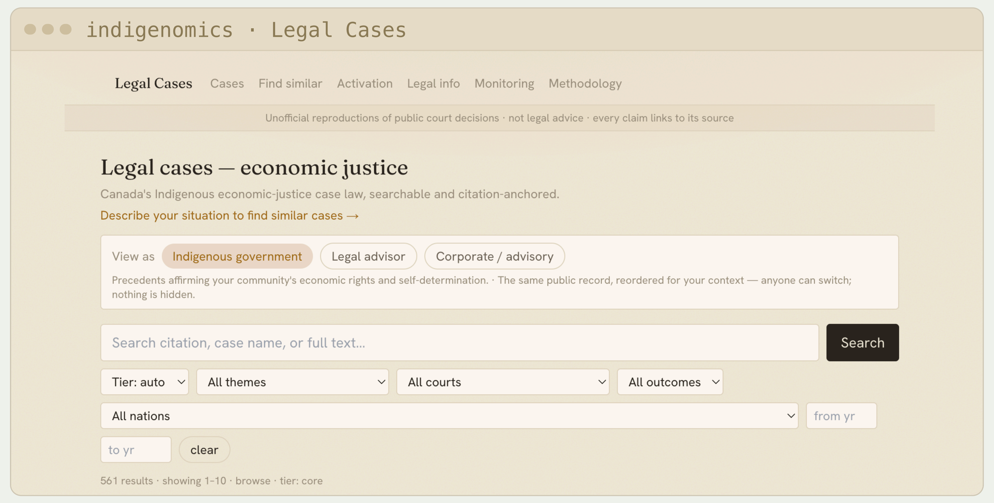
WHO USES IT

- Indigenomics Institute
- Indigenous organizations
- Corporate partners
- Researchers & legal advisors



RAP Index
- 106 commitments · 100 organizations
- 63% average progress tracked



Legal Cases
- 561 economic-justice decisions
- citation-anchored, role-based views

ARCHITECTURE — SERVERLESS ON AWS

CloudFront edge → **Next.js** on Lambda → **Textract** OCR → **Bedrock** extraction → **DynamoDB** ×6 tables

WHAT THE INSTITUTE CAN NOW DO

- See every tracked commitment and its progress in one view
- Get a weekly digest of what has gone **overdue**
- Trace any claim back to the published source document
- Read the legal precedent beside the corporate promise

02 HOW IT BECAME RESEARCH

Client problem
reconciliation data scattered & unverifiable

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We built the portal
production platform, shipped on AWS

↓

179 pull requests
30 of them ours — real, merged, AI-reviewed

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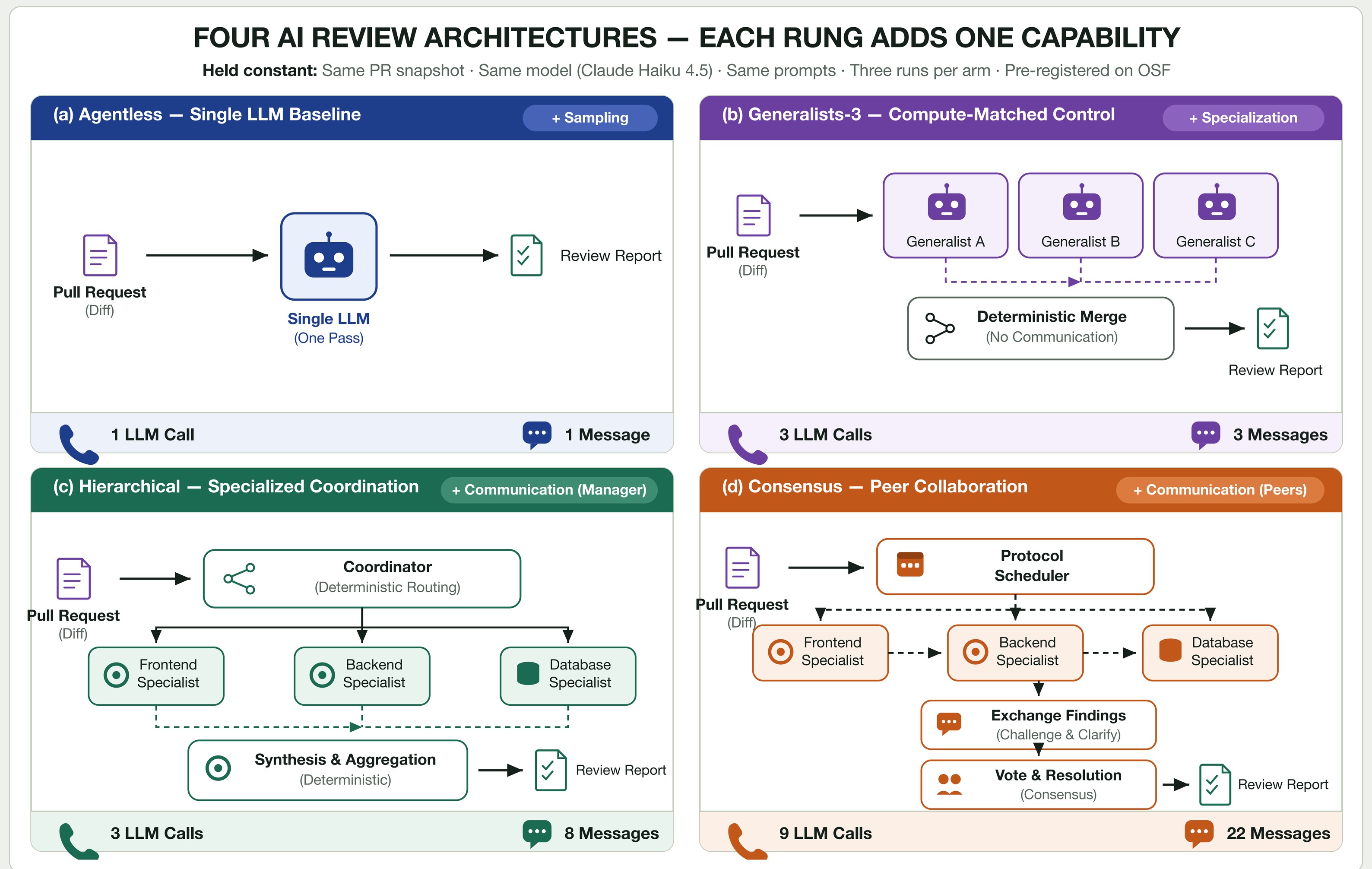
Benchmark dataset
E1 Qodo · E2 SWE-PRBench · E3 our portal — 2,415 reviews

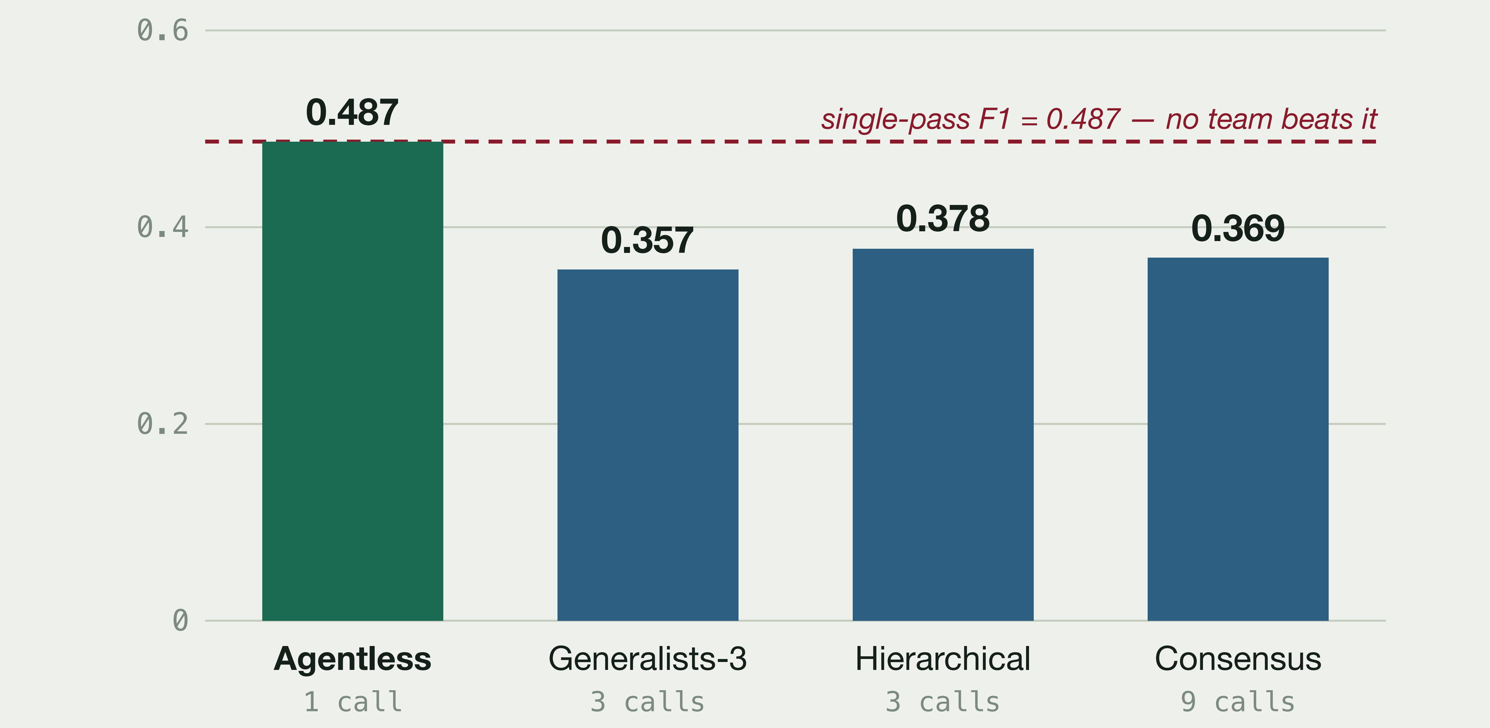
↓

Four AI review architectures
each rung adds exactly one capability

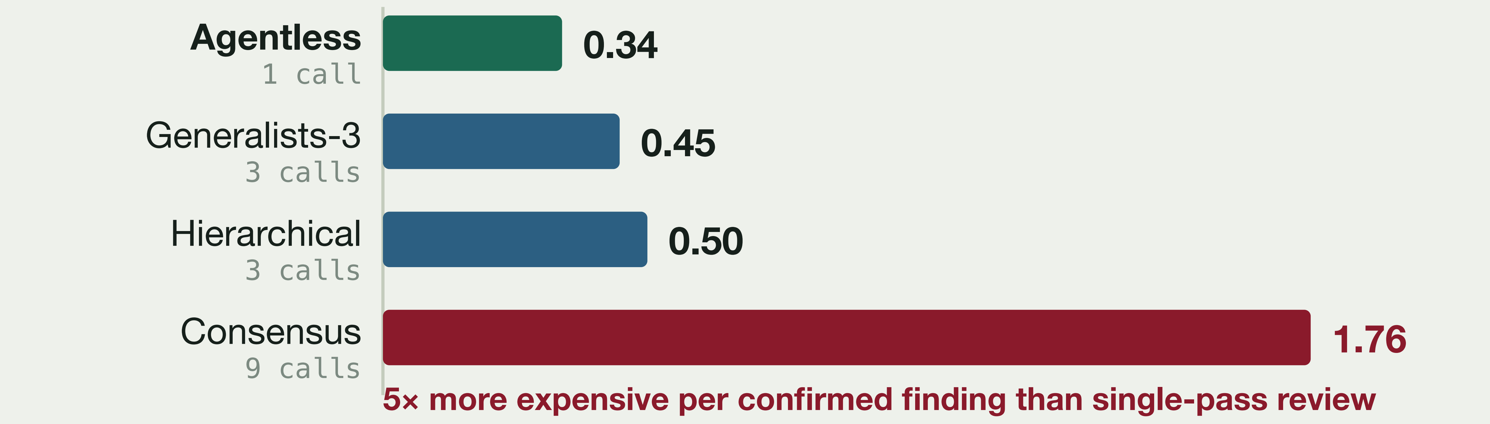


03 WHAT WE FOUND · E1 · QODO, 99 PRS

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling (p=0.73); peer debate adds nothing over a manager (p=0.12).



Extra agents buy recall at a price — up to **5×** the cost per confirmed finding.



Bounded. Recall plateaus at ≈**0.83**; on our own unlabeled code, LLM judges disagree (**κ=0.03**).

WHERE THE SIGNAL LIVES

Agreement recovers **universal functional bugs** far better than project-specific conventions — which is why the ceiling exists.

43% functional-bug recall at 3 independent families agreeing

18% rule-violation recall at the same depth

Pair decorrelated review with a deterministic linter.

PROJECT IMPACT

- **Production platform delivered to the Indigenomics Institute**
- **Reconciliation commitments made searchable and verifiable**
- **Indigenous legal precedent explorer, citation-anchored**

RESEARCH IMPACT

- **First compute-matched comparison of review architectures**
- **Independence beats more agents — decorrelate your verifiers**
- **Guidance for anyone deploying AI code review in production**

Code & data rap-review-research

Live portal try the demo

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Pre-registered on OSF (embargoed)